

# Udit Agarwal

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## EDUCATION

**Indiana University Bloomington** - Bloomington, IN, USA

Aug 2024 – May 2026

**Master of Science in Computer Science**

**GPA: 3.82 / 4.0**

- *Relevant Coursework: Advanced Database Concepts, Applied Algorithms, Applied Machine Learning, Computer Networks, Engineering Cloud Computing, Fundamentals & Applications of LLMs, Software Engineering*

**Kalinga Institute of Industrial Technology (KIIT University)** - Bhubaneswar, OD, India

Aug 2020 – May 2024

**Bachelor of Technology in Computer Science & Engineering**

**GPA: 8.85 / 10.0**

## TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, C/C++, SQL
- **Full-Stack:** React, Next.js, Node.js, REST APIs
- **Data & Databases:** PostgreSQL, MySQL, MongoDB, Redis, Pandas, NumPy
- **AI/ML:** Scikit-learn, TensorFlow, NLP, Transformers (BERT, DeBERTa), LLM APIs
- **Data Engineering:** ETL Pipelines, Data Modeling, Batch Processing
- **Analytics & Tools:** Tableau, Matplotlib, Excel
- **Systems & Cloud:** AWS, Docker, Git, System Design, Distributed Systems

## EXPERIENCE

**Global Health Impact Project**

Feb 2026 – Present

*Software Engineer*

*Indiana University - Bloomington, IN*

- Developing a scalable full-stack platform to analyze pharmaceutical interventions across populations
- Building scalable APIs and data processing pipelines to support forecasting and analytics workflows
- Contributing to models evaluating treatment coverage and health outcomes using real-world datasets
- Collaborating with cross-functional teams to translate research into production-ready solutions (volunteer role)

**University Information Technology Services (UITS)**

Aug 2025 – Present

*Consultant – IT Services*

*Indiana University - Bloomington, IN*

- Resolved 100+ technical issues weekly across enterprise systems, improving system reliability and user experience
- Applied structured troubleshooting and root-cause analysis to reduce recurring system issues
- Managed identity and access systems ensuring secure authentication and seamless user access
- Documented workflows and support patterns to improve operational efficiency and service delivery

## PROJECTS

**AI Mock Interview Platform** | *Next.js, Firebase, Node.js, Vapi AI, Gemini API*

- Built a full-stack AI platform to simulate real-time interview scenarios using voice agents and LLM APIs
- Designed backend workflows for dynamic question generation and structured feedback evaluation
- Developed data pipelines to capture user responses and generate real-time performance insights

**Screen Recording & Video Sharing Platform** | *Next.js, TypeScript, Bunny.net, Xata, Drizzle ORM*

- Developed a full-stack platform for screen recording, video uploads, and secure sharing
- Implemented scalable backend architecture with CDN-based media delivery
- Designed features including authentication, privacy controls, and transcript generation

**Automated Secrets Scanner** | *Python, FastAPI, PostgreSQL*

- Built a backend system to detect hardcoded secrets using pattern matching and entropy-based analysis
- Designed scalable scanning pipelines to analyze large codebases and identify vulnerabilities
- Developed APIs and reporting mechanisms to surface security risks and improve code safety

**Business Analytics Dashboard** | *React, Node.js, PostgreSQL, Tableau*

- Built a decision-support system to track KPIs and analyze business performance across datasets
- Designed SQL queries and aggregation pipelines for trend analysis and forecasting
- Implemented dashboards enabling scenario-based decision making and performance evaluation

**Brain Tumor Classification** | *Python, TensorFlow, CNNs*

- Developed deep learning models achieving 99.84% accuracy across 7,000+ MRI scans
- Benchmarked architectures (CNN, VGG16, InceptionV3, EfficientNet) to optimize performance and inference efficiency
- Co-authored [IEEE publication](#) on medical image classification using deep neural networks